TRACKSHIELD:

AN INTELLIGENT AI-IOT SYSTEM FOR RAILWAY TRACK HEALTH AND INTRUSION DETECTION

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Abstract

Railway transportation requires continuous monitoring to ensure the safety of passengers, railway infrastructure, and surrounding areas. Track damage, abnormal vibrations, and unauthorized intrusion onto railway tracks can lead to serious accidents and operational disruptions. This project proposes TrackShield, an intelligent AI-IoT based railway safety monitoring system that combines track condition monitoring with real-time intrusion detection. The system uses an ESP32 microcontroller to collect data from sensors such as an accelerometer and vibration sensor for identifying abnormal track conditions. A HuskyLens AI camera is integrated to detect and recognize people or objects entering the restricted railway track area. When abnormal conditions or unauthorized intrusion are detected, the system generates an immediate warning through a buzzer and display and can transmit alerts through an IoT platform. The proposed system is designed as a low-cost prototype that demonstrates continuous railway safety monitoring and early warning capabilities. By integrating sensor-based monitoring, AI-based visual detection, and Railway transportation requires continuous monitoring to ensure the safety of passengers, railway infrastructure, and surrounding areas. Track damage, abnormal vibrations, and unauthorized intrusion onto railway tracks can lead to serious accidents and operational disruptions. This project proposes TrackShield, an intelligent AI-IoT based railway safety monitoring system that combines track condition monitoring with real-time intrusion detection. The system uses an ESP32 microcontroller to collect data from sensors such as an accelerometer and IoT communication, TrackShield aims to provide a reliable and efficient approach to improving railway safety and reducing the risk of accidents caused by track abnormalities and unauthorized access.

Keywords: Railway Safety; HuskyLens; Intrusion Detection; IoT; Track Monitoring.